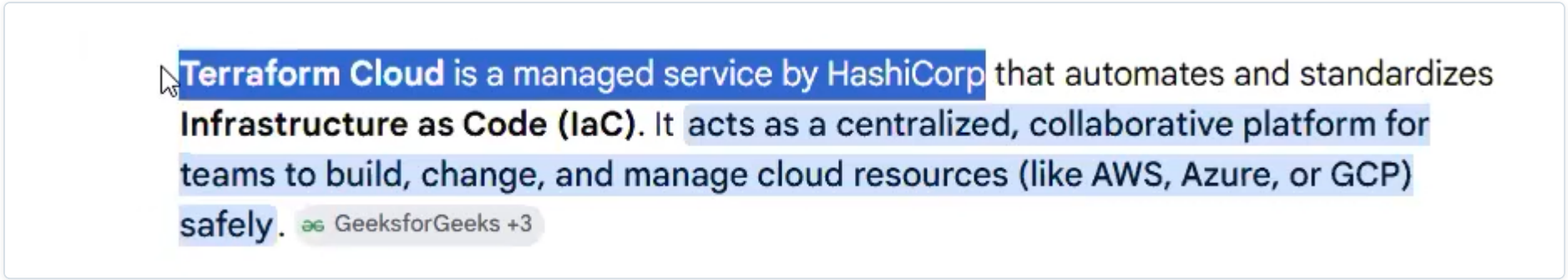
Screenshots — Terraform Cloud (HCP Terraform)

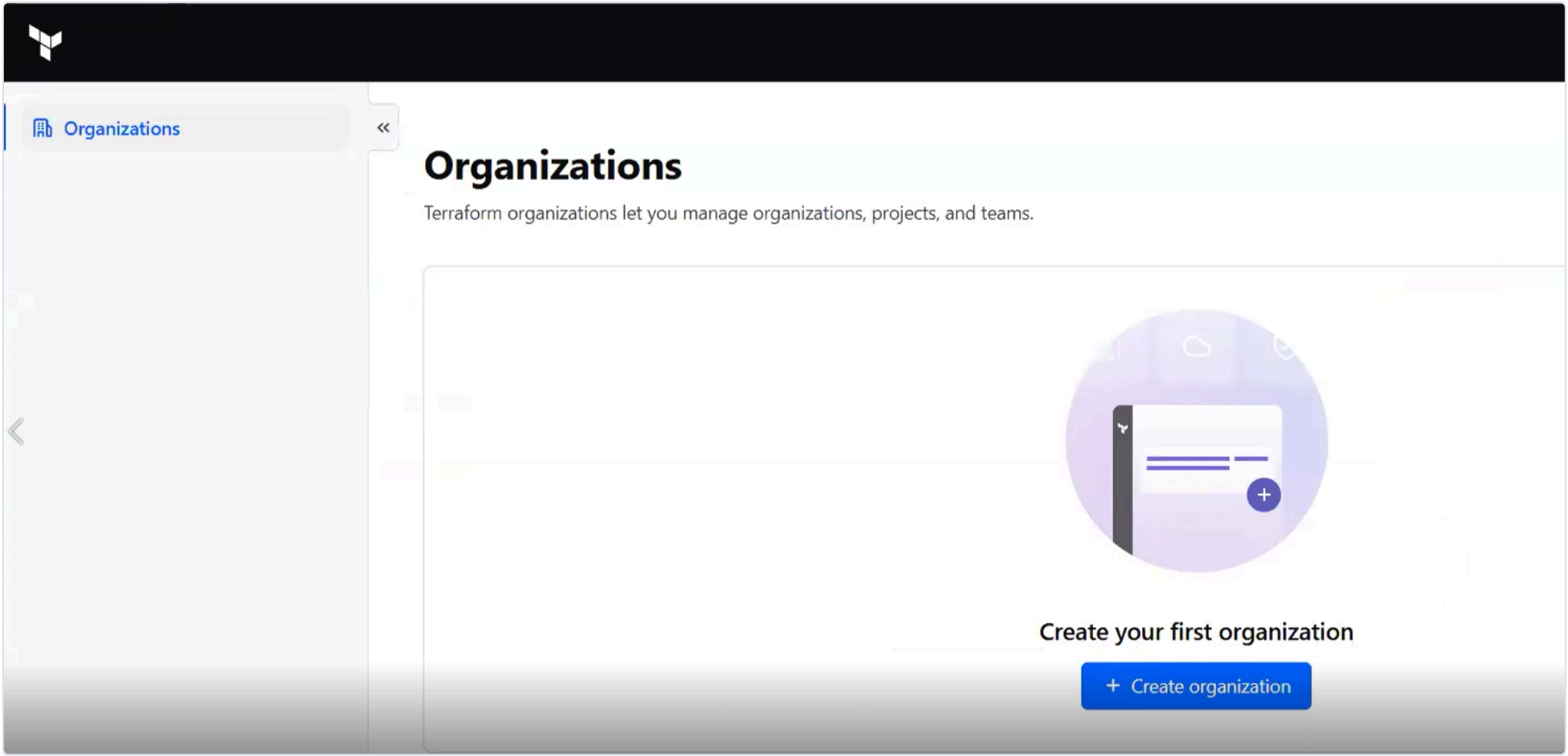
Lecture 11 · 28 shots, chronological · source stills from the session (2026-10-08 19:46 – 22:51)

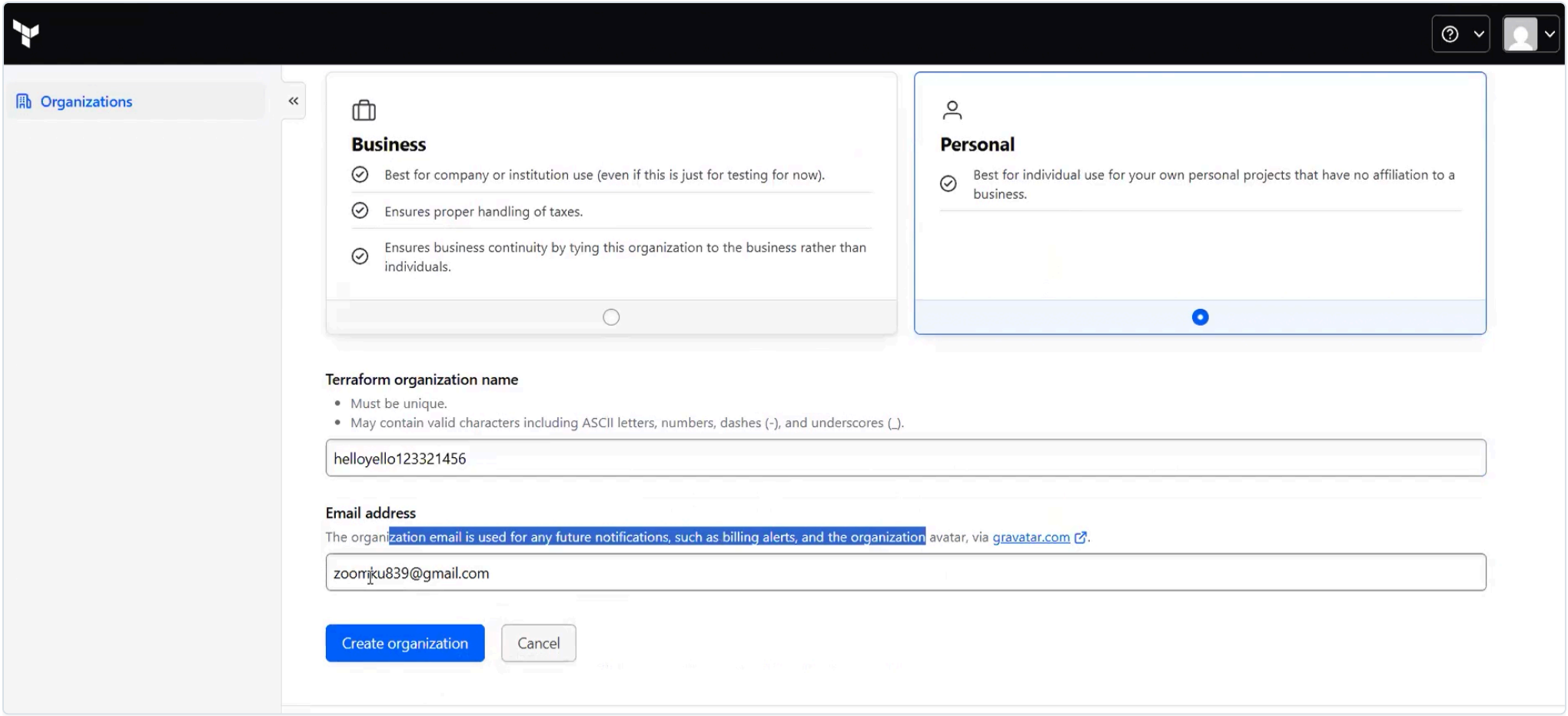
01 / 28 Definition — Terraform Cloud is a managed service by HashiCorp; centralized, collaborative platform for IaC 19:46:55

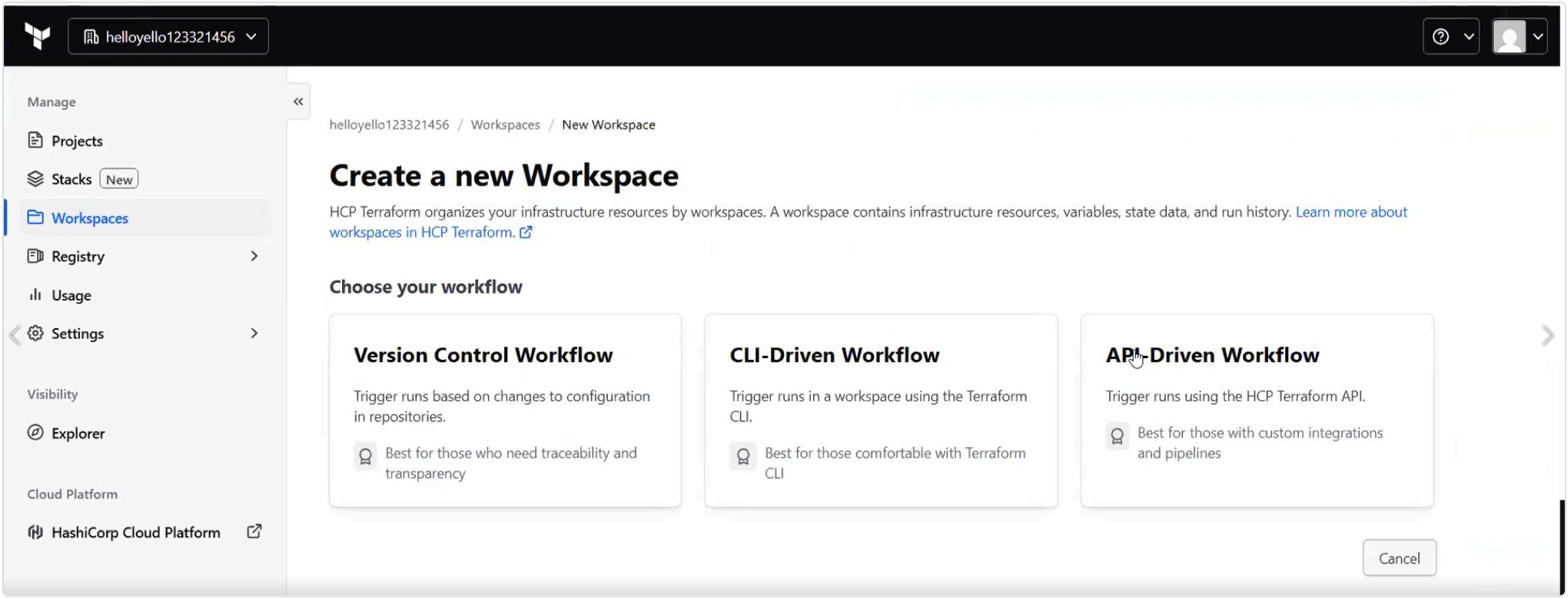


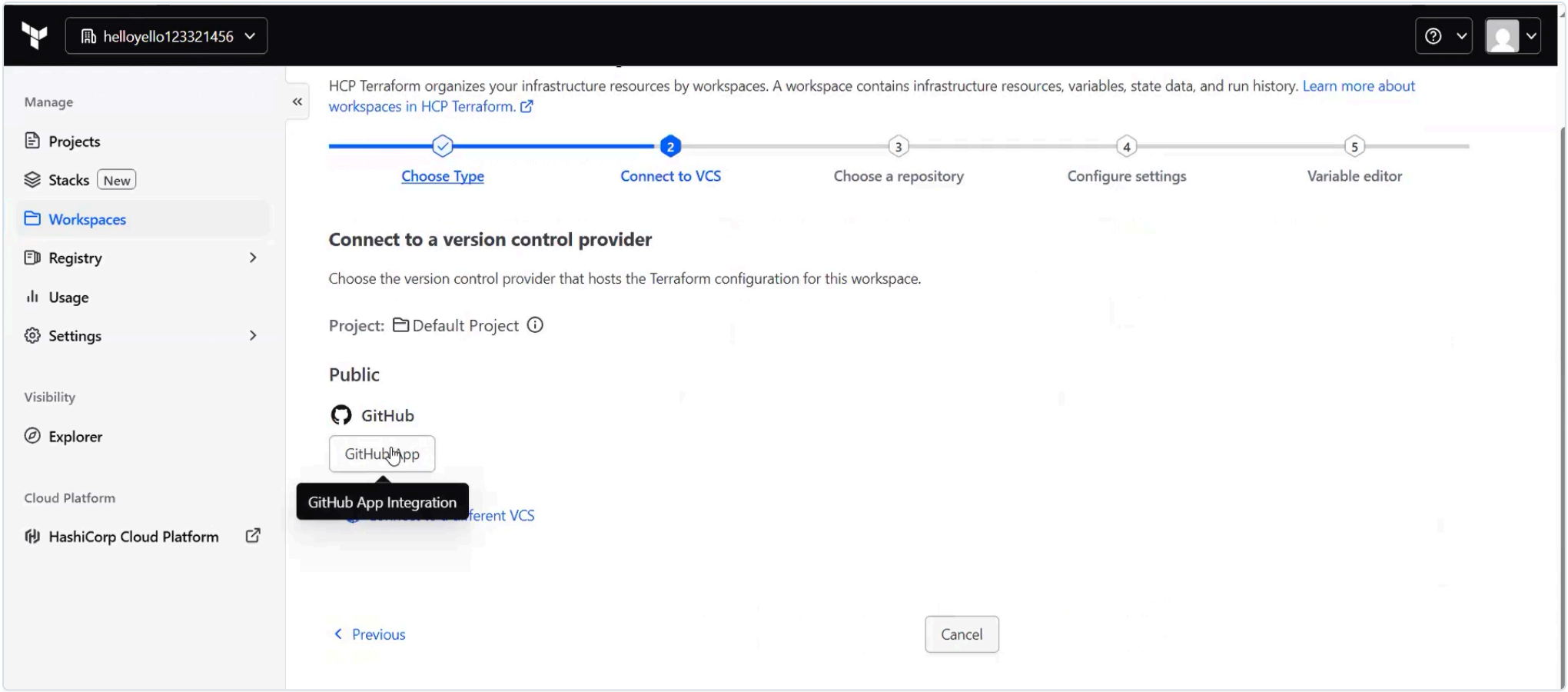
Key capabilities include:

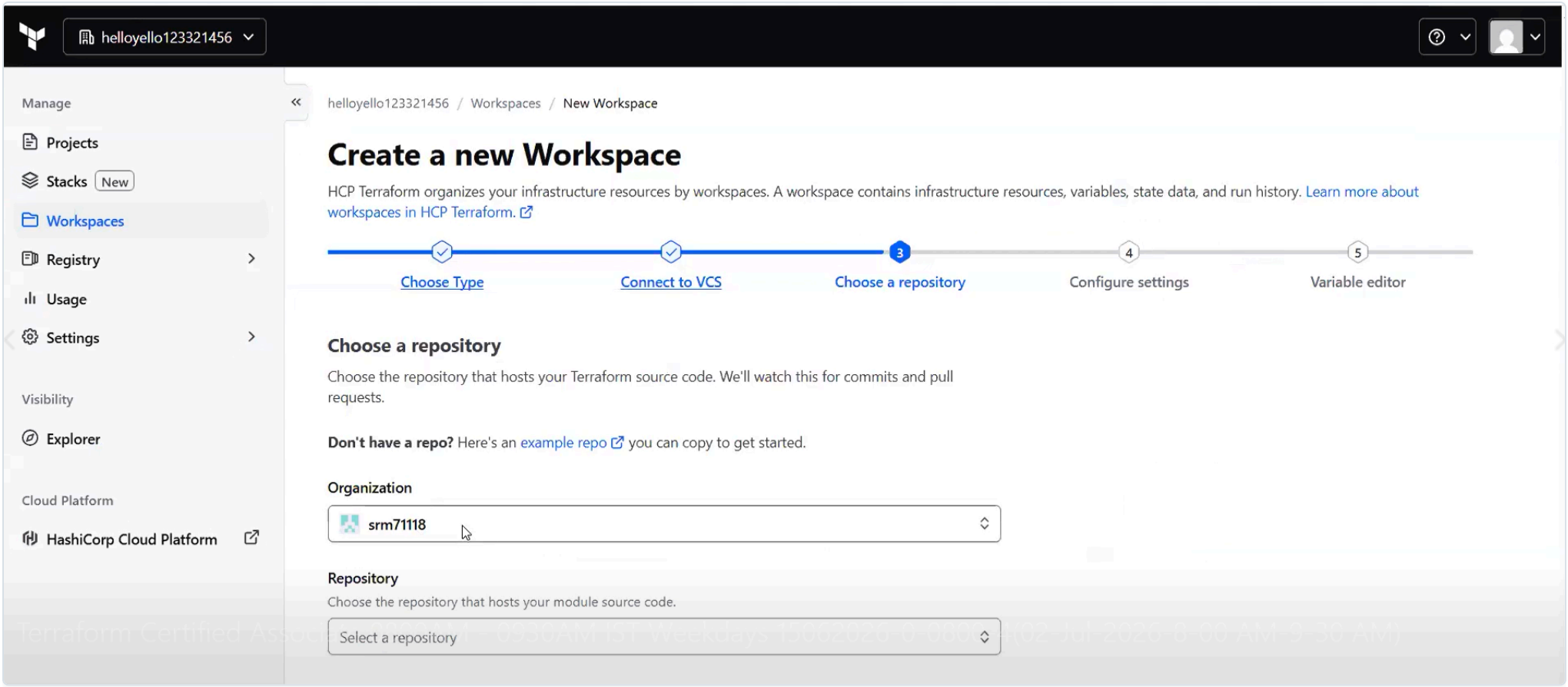
- **Remote State Management:** Safely stores infrastructure state files in the cloud with automatic versioning and state locking, eliminating the risks of conflicts when multiple team members make changes. HashiCorp +2
- **Collaboration & VCS Integration:** Connects directly with Git repositories (like GitHub, GitLab, and Bitbucket) to automatically run Terraform plans when code is pushed or when pull requests are created. YouTube · env zero +2
- **Automated & Consistent Runs:** Executes all infrastructure provisioning in a secure, isolated cloud environment, removing the need to run Terraform from individual engineers' laptops. GeeksforGeeks +1
- **Governance and Policy-as-Code:** Uses policy engines to enforce compliance standards and automatically estimates the costs of resources before they are built. GeeksforGeeks +1
- **Role-Based Access Control (RBAC):** Allows organizations to restrict permissions, ensuring that only authorized users can approve changes to production environments. HashiCorp +1

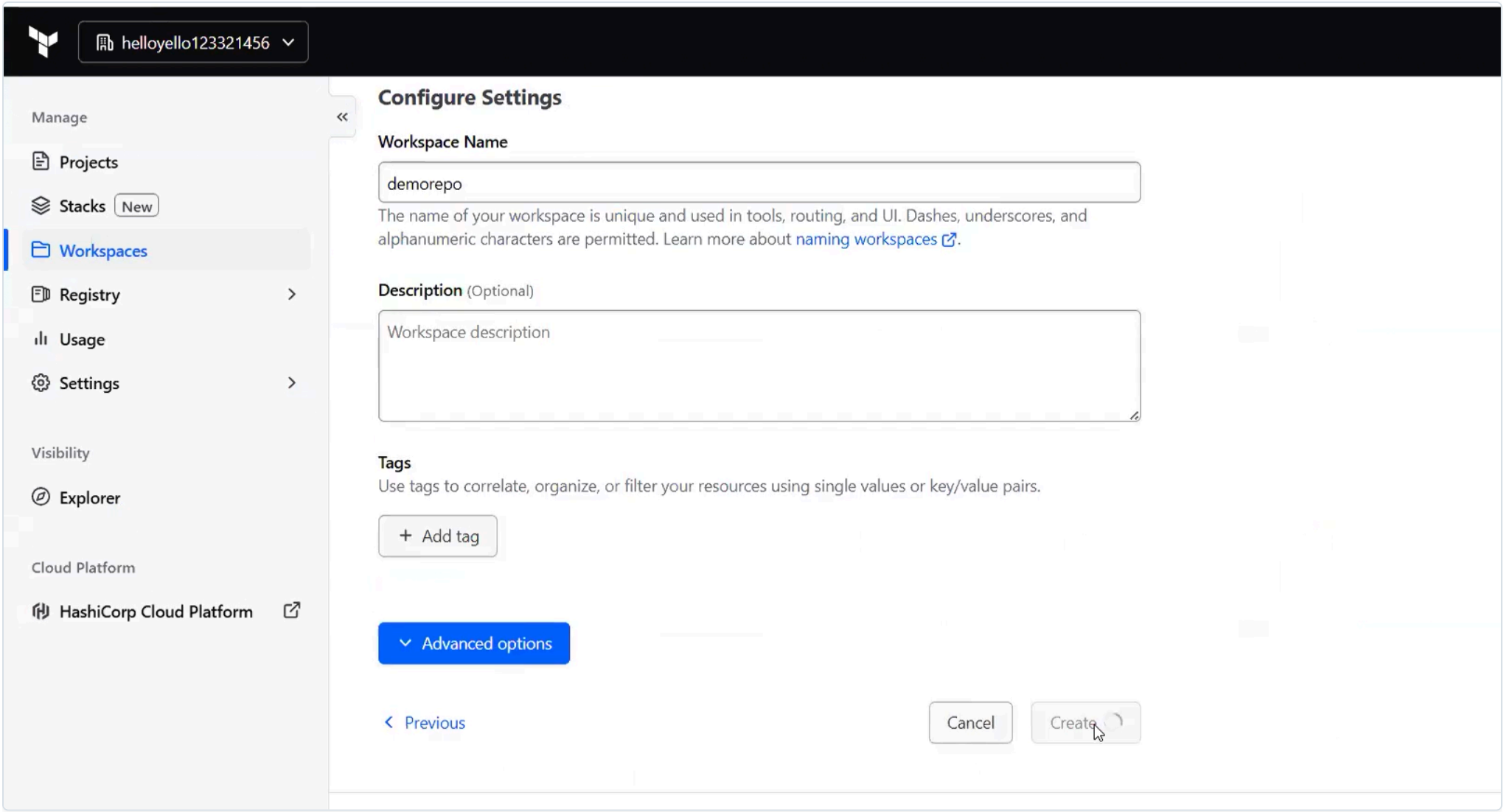

Organizations
Organizations
Terraform organizations let you manage organizations, projects, and teams.
Create your first organization
+ Create organization

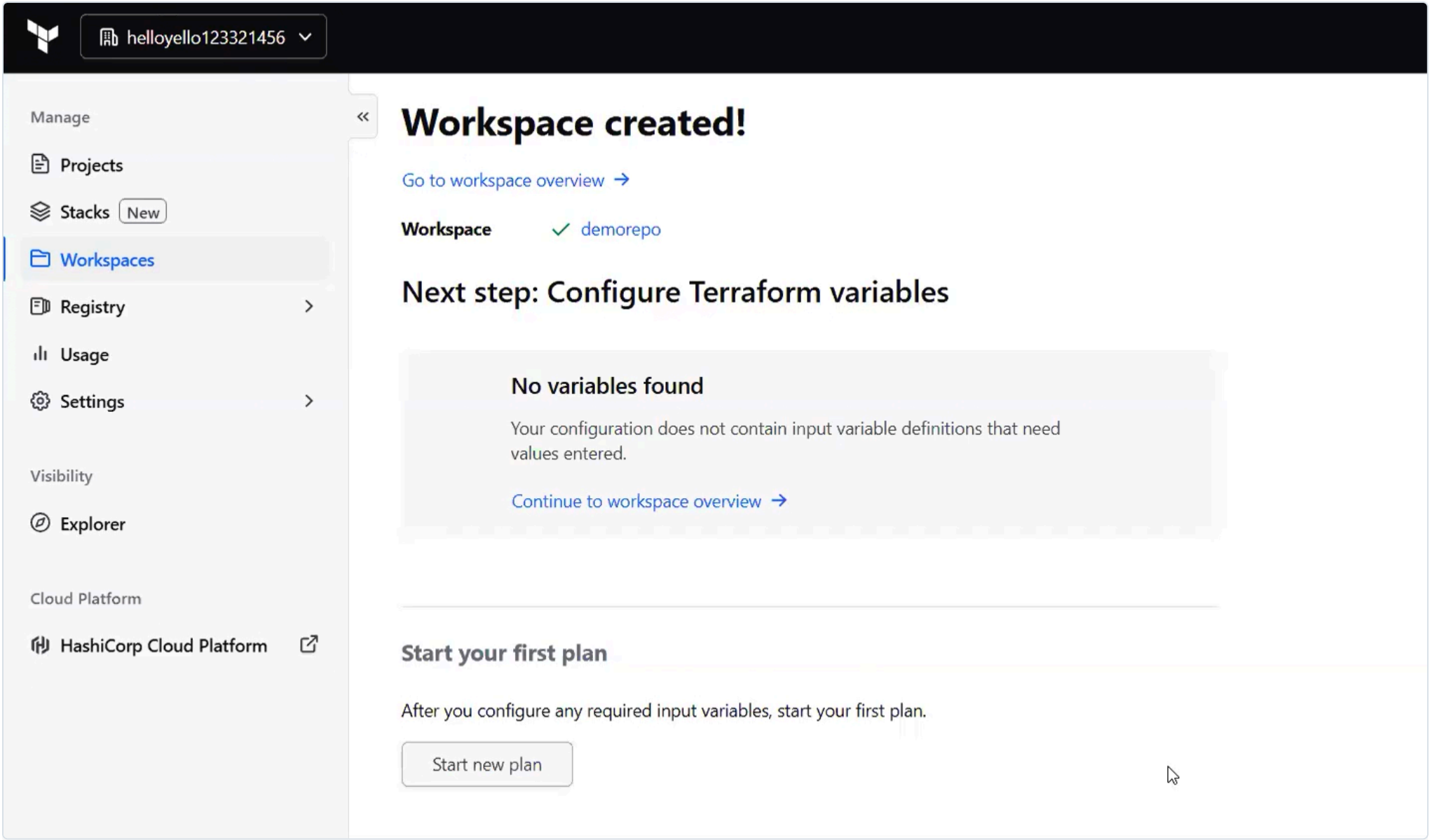

Organizations
Business
Best for company or institution use (even if this is just for testing for now).
Ensures proper handling of taxes.
Ensures business continuity by tying this organization to the business rather than individuals.
Personal
Best for individual use for your own personal projects that have no affiliation to a business.
Terraform organization name
Must be unique.
May contain valid characters including ASCII letters, numbers, dashes (-), and underscores (_).
helloyello123321456
Email address
The organization email is used for any future notifications, such as billing alerts, and the organization avatar, via gravatar.com.
zoomku839@gmail.com
Create organization
Cancel

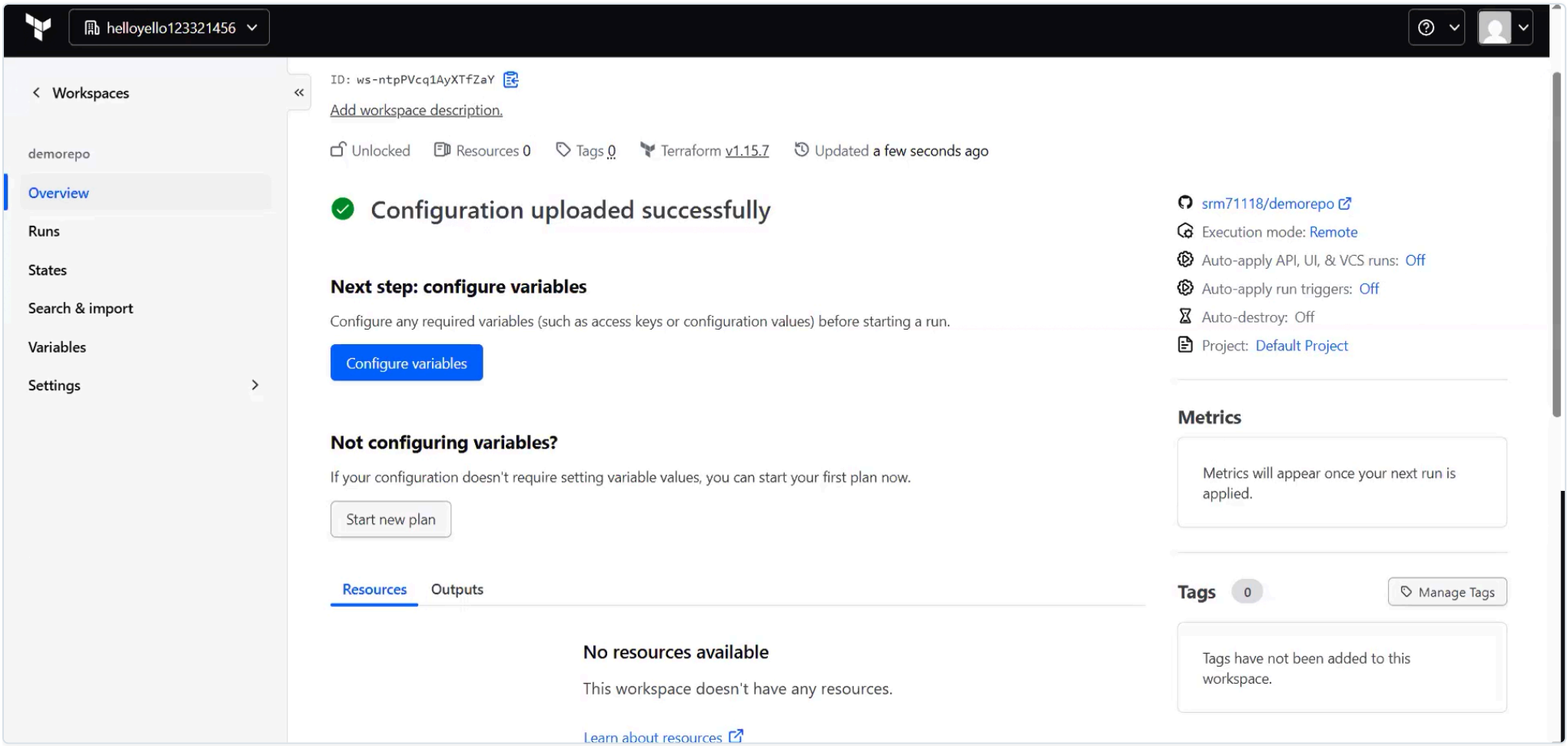

helloyello123321456
Manage
Projects
Stacks New
Workspaces
Registry
Usage
Settings
Visibility
Explorer
Cloud Platform
HashiCorp Cloud Platform
helloyello123321456 / Workspaces / New Workspace
Create a new Workspace
HCP Terraform organizes your infrastructure resources by workspaces. A workspace contains infrastructure resources, variables, state data, and run history. Learn more about workspaces in HCP Terraform.
Choose your workflow
Version Control Workflow
Trigger runs based on changes to configuration in repositories.
Best for those who need traceability and transparency
CLI-Driven Workflow
Trigger runs in a workspace using the Terraform CLI.
Best for those comfortable with Terraform CLI
API-Driven Workflow
Trigger runs using the HCP Terraform API.
Best for those with custom integrations and pipelines
Cancel

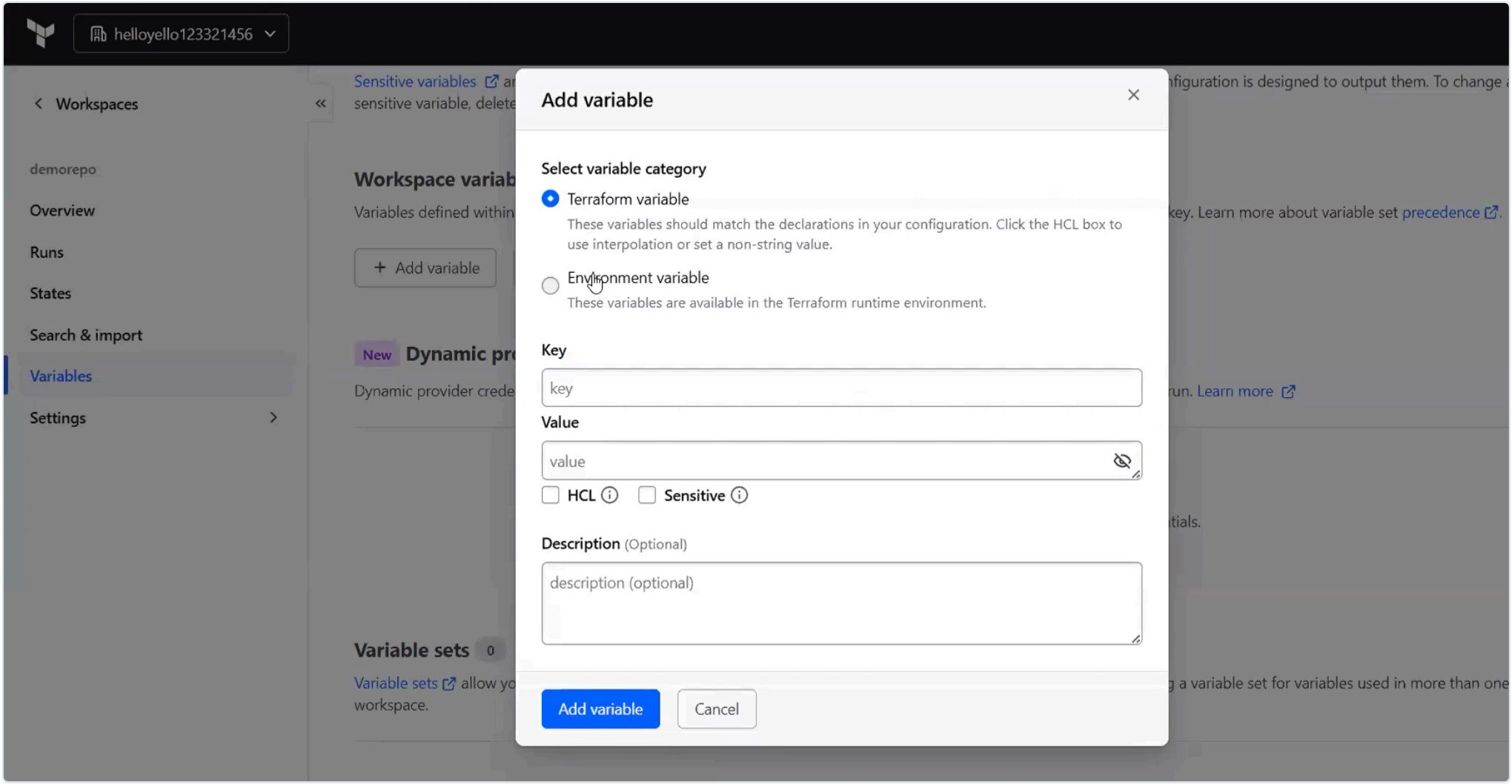

helloyello123321456
Manage
Projects
Stacks New
Workspaces
Registry
Usage
Settings
Visibility
Explorer
Cloud Platform
HashiCorp Cloud Platform
HCP Terraform organizes your infrastructure resources by workspaces. A workspace contains infrastructure resources, variables, state data, and run history. Learn more about workspaces in HCP Terraform.
Choose Type
2
Connect to VCS
3
Choose a repository
4
Configure settings
5
Variable editor
Connect to a version control provider
Choose the version control provider that hosts the Terraform configuration for this workspace.
Project: Default Project
Public
GitHub
GitHub App
GitHub App Integration
ferent VCS
Previous
Cancel

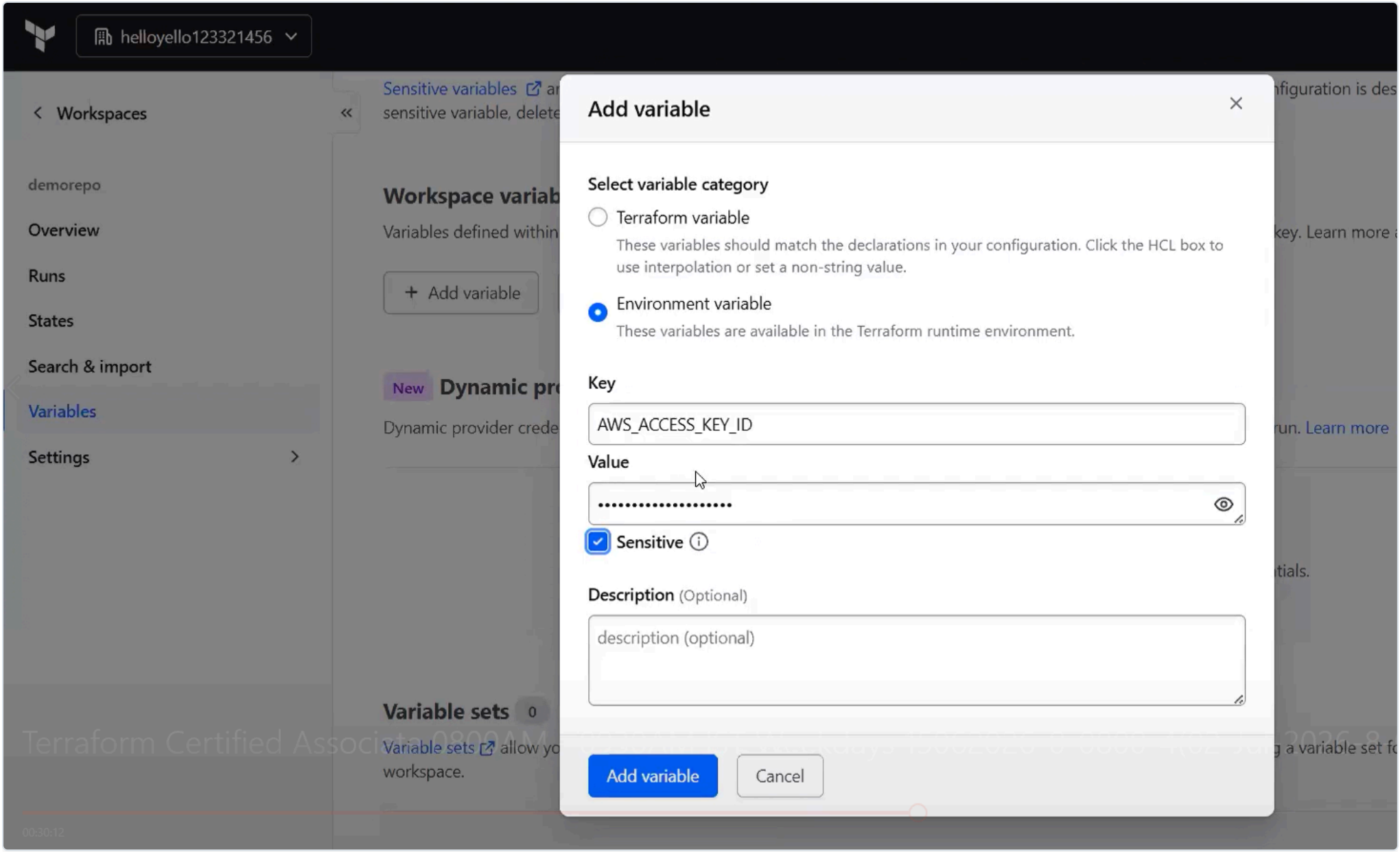

helloyello123321456
Manage
Projects
Stacks New
Workspaces
Registry
Usage
Settings
Visibility
Explorer
Cloud Platform
HashiCorp Cloud Platform
helloyello123321456 / Workspaces / New Workspace
Create a new Workspace
HCP Terraform organizes your infrastructure resources by workspaces. A workspace contains infrastructure resources, variables, state data, and run history. Learn more about workspaces in HCP Terraform.
Choose Type
Connect to VCS
3
Choose a repository
4
Configure settings
5
Variable editor
Choose a repository
Choose the repository that hosts your Terraform source code. We'll watch this for commits and pull requests.
Don't have a repo? Here's an example repo you can copy to get started.
Organization
srm71118
Repository
Choose the repository that hosts your module source code.
Select a repository

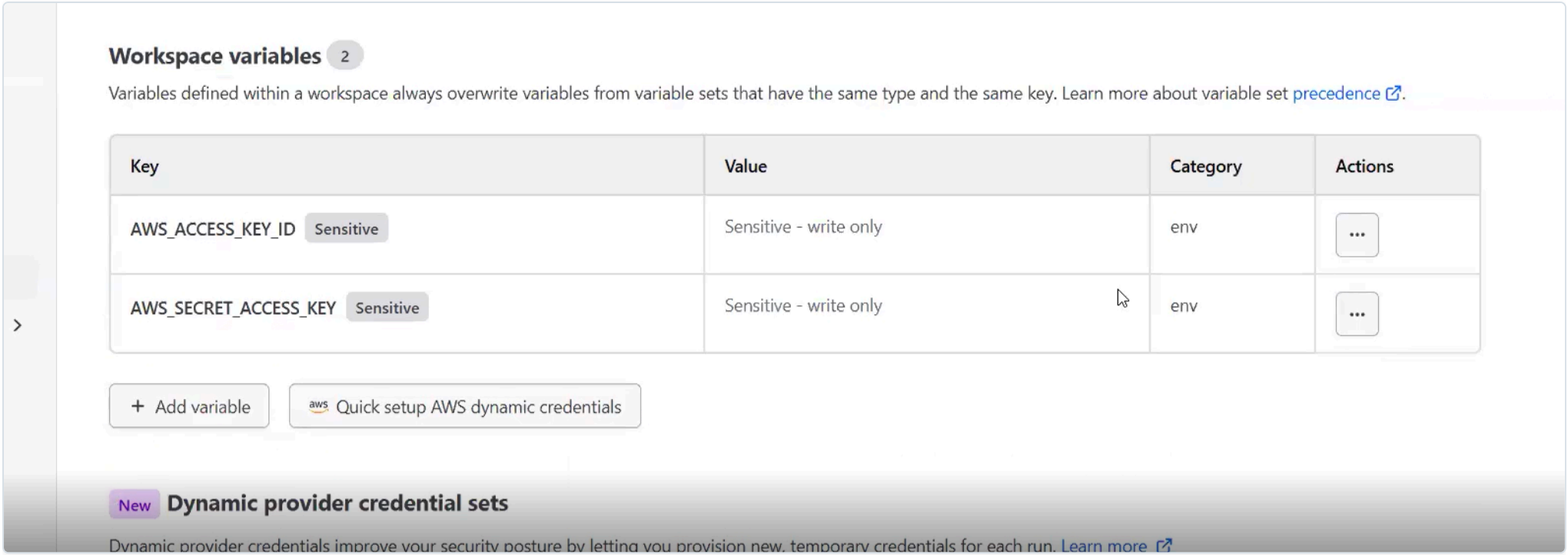

helloyello123321456
Manage
Projects
Stacks New
Workspaces
Registry
Usage
Settings
Visibility
Explorer
Cloud Platform
HashiCorp Cloud Platform
Configure Settings
Workspace Name
demorepo
The name of your workspace is unique and used in tools, routing, and UI. Dashes, underscores, and alphanumeric characters are permitted. Learn more about naming workspaces.
Description (Optional)
Workspace description
Tags
Use tags to correlate, organize, or filter your resources using single values or key/value pairs.
+ Add tag
Advanced options
Previous
Cancel
Create

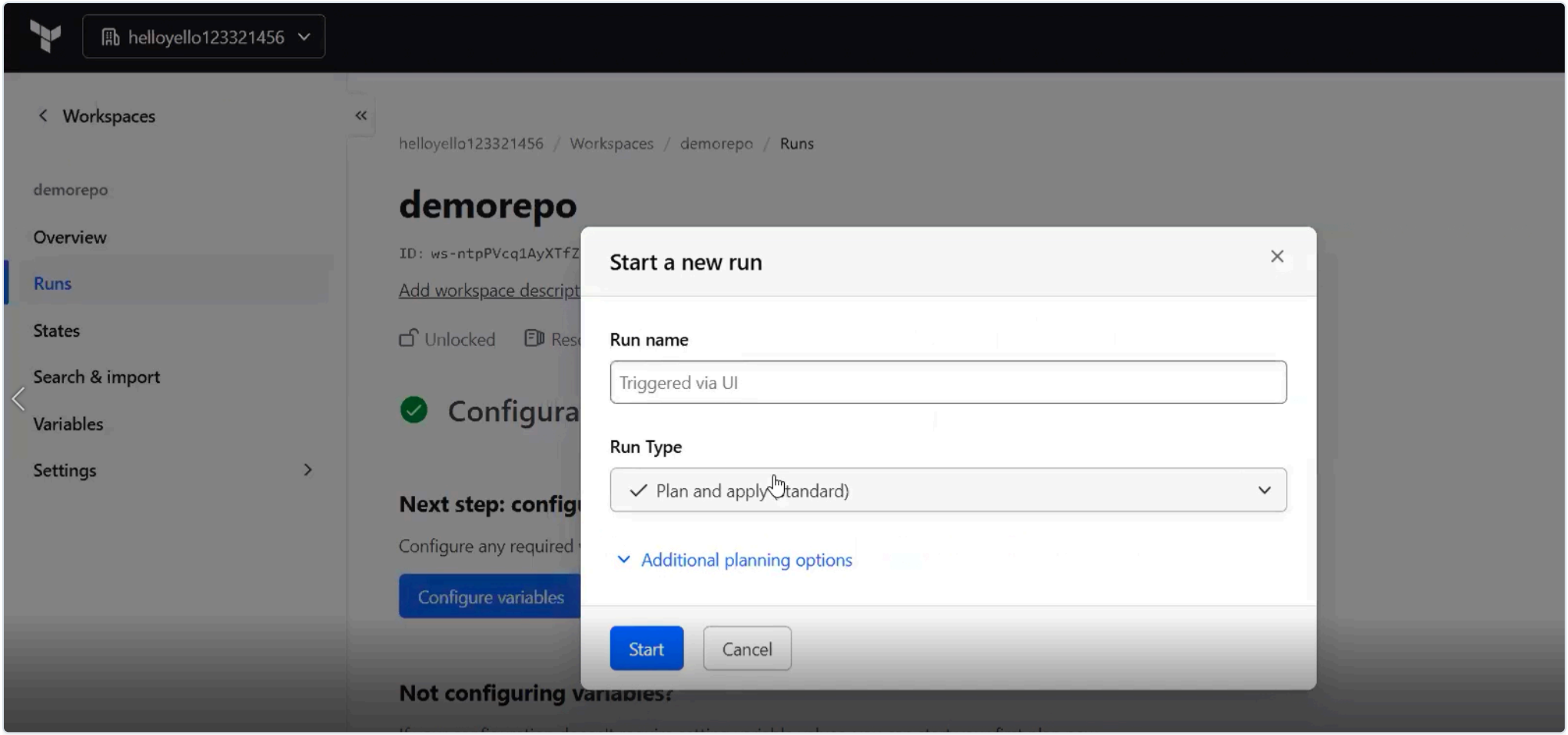

helloyello123321456
Manage
Projects
Stacks New
Workspaces
Registry
Usage
Settings
Visibility
Explorer
Cloud Platform
HashiCorp Cloud Platform
Workspace created!
Go to workspace overview
Workspace
demorepo
Next step: Configure Terraform variables
No variables found
Your configuration does not contain input variable definitions that need values entered.
Continue to workspace overview
Start your first plan
After you configure any required input variables, start your first plan.
Start new plan

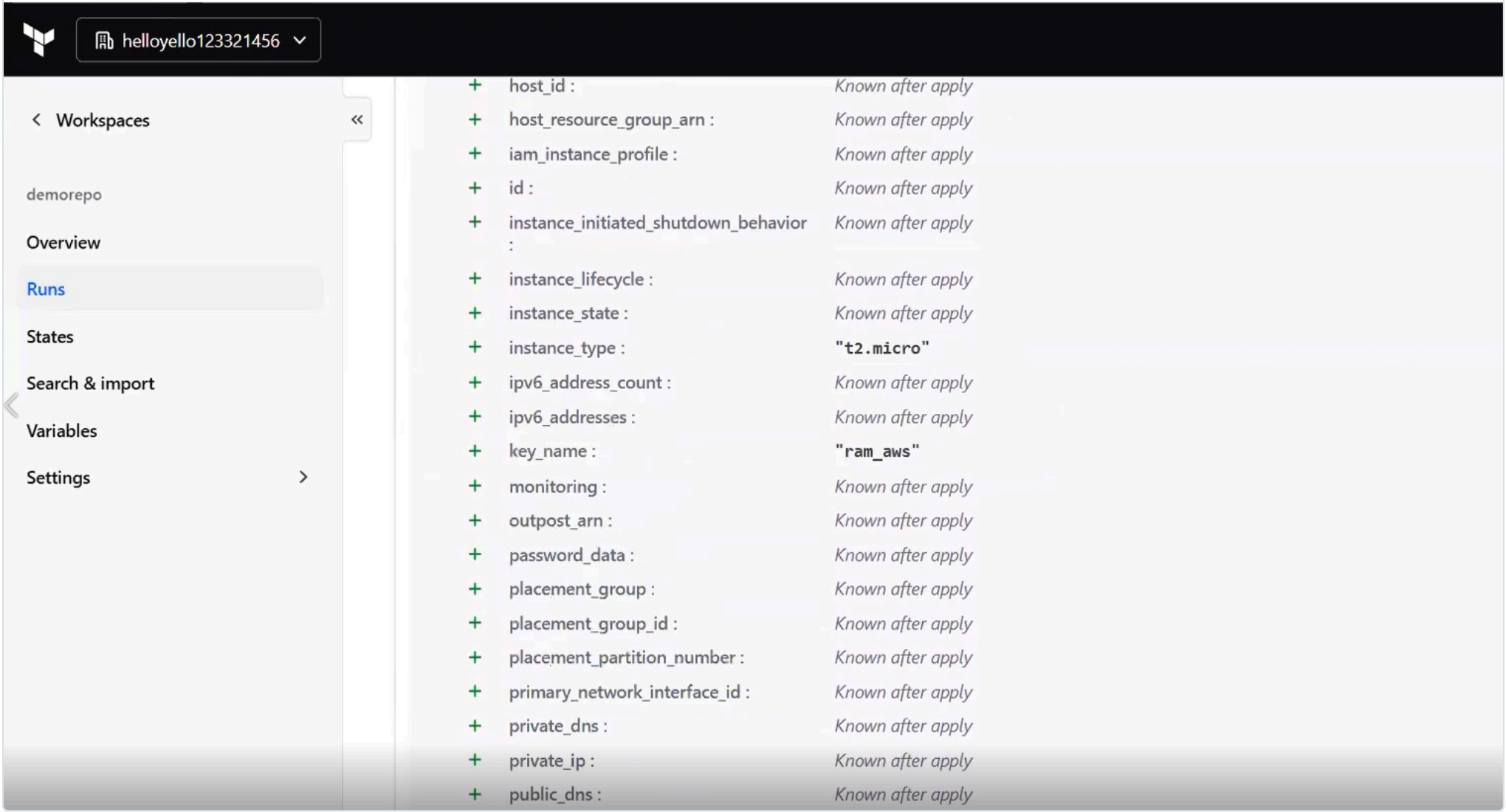

helloyello123321456
Workspaces
demorepo
Overview
Runs
States
Search & import
Variables
Settings
ID: ws-ntpPVcq1AyXTfZaY
Add workspace description.
Unlocked
Resources 0
Tags 0
Terraform v1.15.7
Updated a few seconds ago
Configuration uploaded successfully
Next step: configure variables
Configure any required variables (such as access keys or configuration values) before starting a run.
Configure variables
Not configuring variables?
If your configuration doesn't require setting variable values, you can start your first plan now.
Start new plan
Resources
Outputs
No resources available
This workspace doesn't have any resources.
Learn about resources
srm71118/demorepo
Execution mode: Remote
Auto-apply API, UI, & VCS runs: Off
Auto-apply run triggers: Off
Auto-destroy: Off
Project: Default Project
Metrics
Metrics will appear once your next run is applied.
Tags 0
Manage Tags
Tags have not been added to this workspace.


helloyello123321456
Workspaces
demorepo
Overview
Runs
States
Search & import
Variables
Settings
Sensitive variables
sensitive variable, delete
figuration is designed to output them. To change
Workspace variab
Variables defined within
key. Learn more about variable set precedence
+ Add variable
New Dynamic pr
Dynamic provider crede
run. Learn more
tials.
Variable sets 0
Variable sets allow yo
workspace.
g a variable set for variables used in more than one
Add variable
Select variable category
Terraform variable
These variables should match the declarations in your configuration. Click the HCL box to use interpolation or set a non-string value.
Environment variable
These variables are available in the Terraform runtime environment.
Key
key
Value
value
HCL
Sensitive
Description (Optional)
description (optional)
Add variable
Cancel

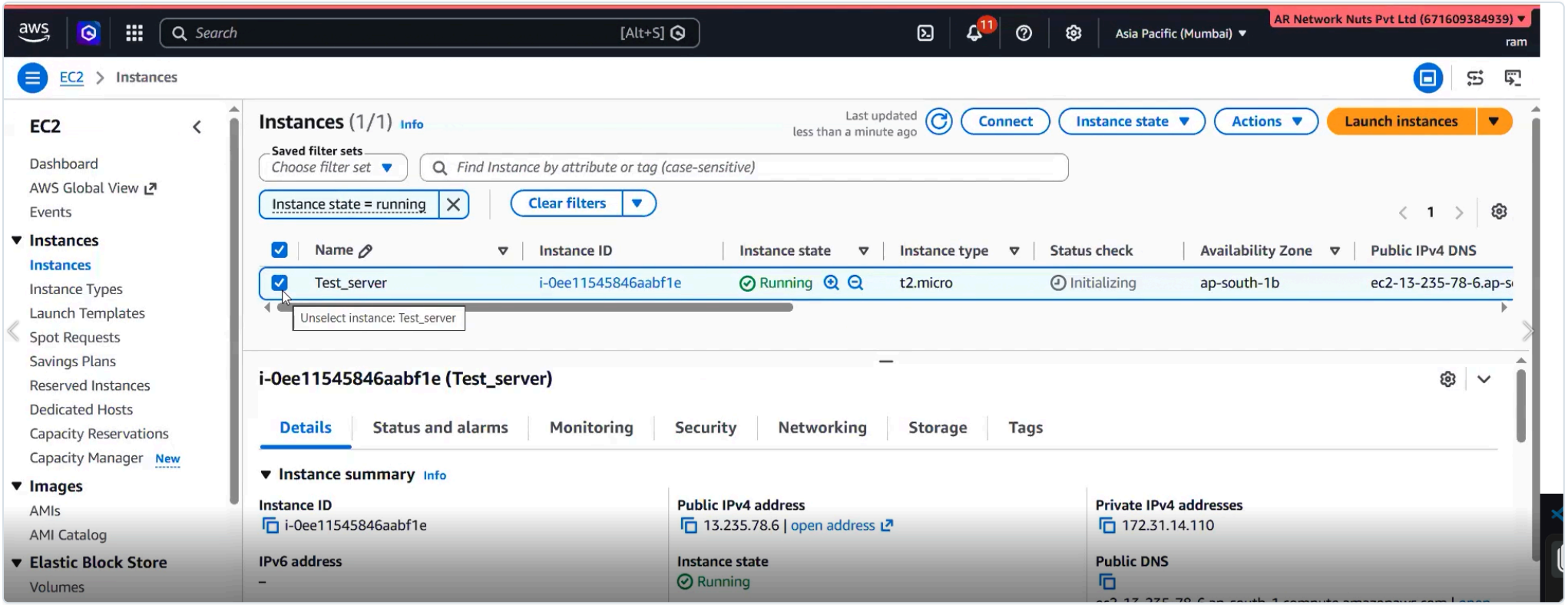

helloyello123321456
Workspaces
demorepo
Overview
Runs
States
Search & import
Variables
Settings
Sensitive variables
sensitive variable, delete
Workspace variab
Variables defined within
+ Add variable
New Dynamic pr
Dynamic provider crede
Variable sets 0
Variable sets allow yo
workspace.
Add variable
Select variable category
Terraform variable
These variables should match the declarations in your configuration. Click the HCL box to use interpolation or set a non-string value.
Environment variable
These variables are available in the Terraform runtime environment.
Key
AWS_ACCESS_KEY_ID
Value
Sensitive
Description (Optional)
description (optional)
Add variable
Cancel

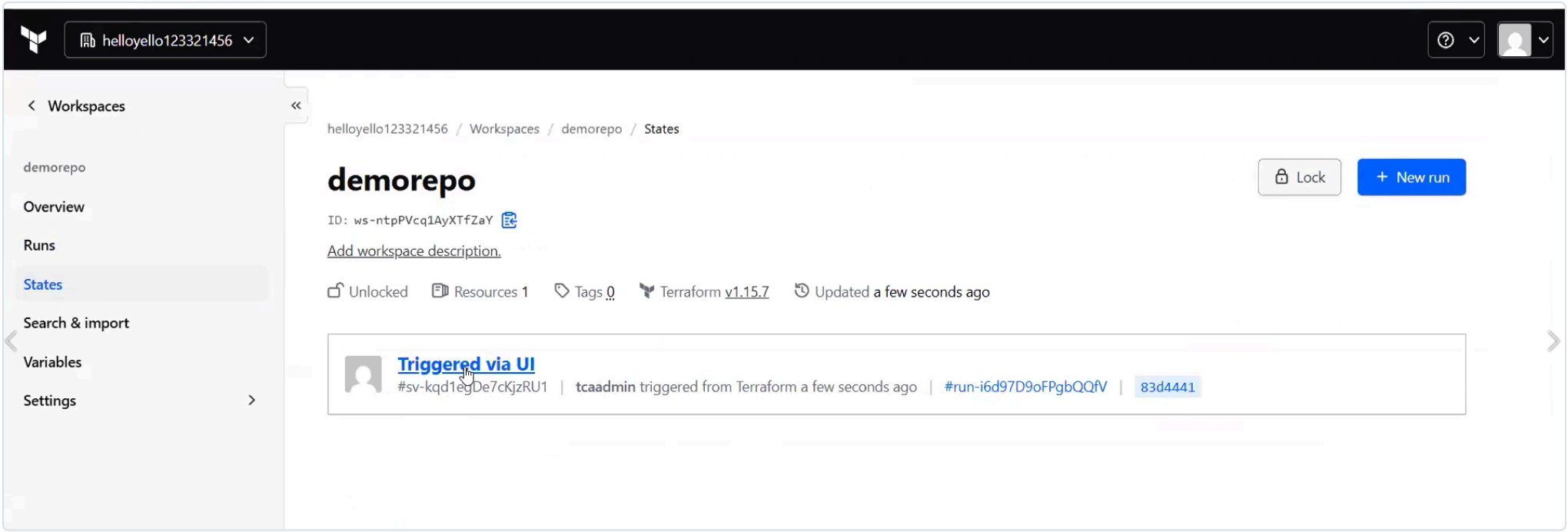

Workspace variables 2
Variables defined within a workspace always overwrite variables from variable sets that have the same type and the same key. Learn more about variable set precedence.
Key
Value
Category
Actions
AWS_ACCESS_KEY_ID Sensitive
Sensitive - write only
env
AWS_SECRET_ACCESS_KEY Sensitive
Sensitive - write only
env
+ Add variable
Quick setup AWS dynamic credentials
New Dynamic provider credential sets
Dynamic provider credentials improve your security posture by letting you provision new, temporary credentials for each run. Learn more

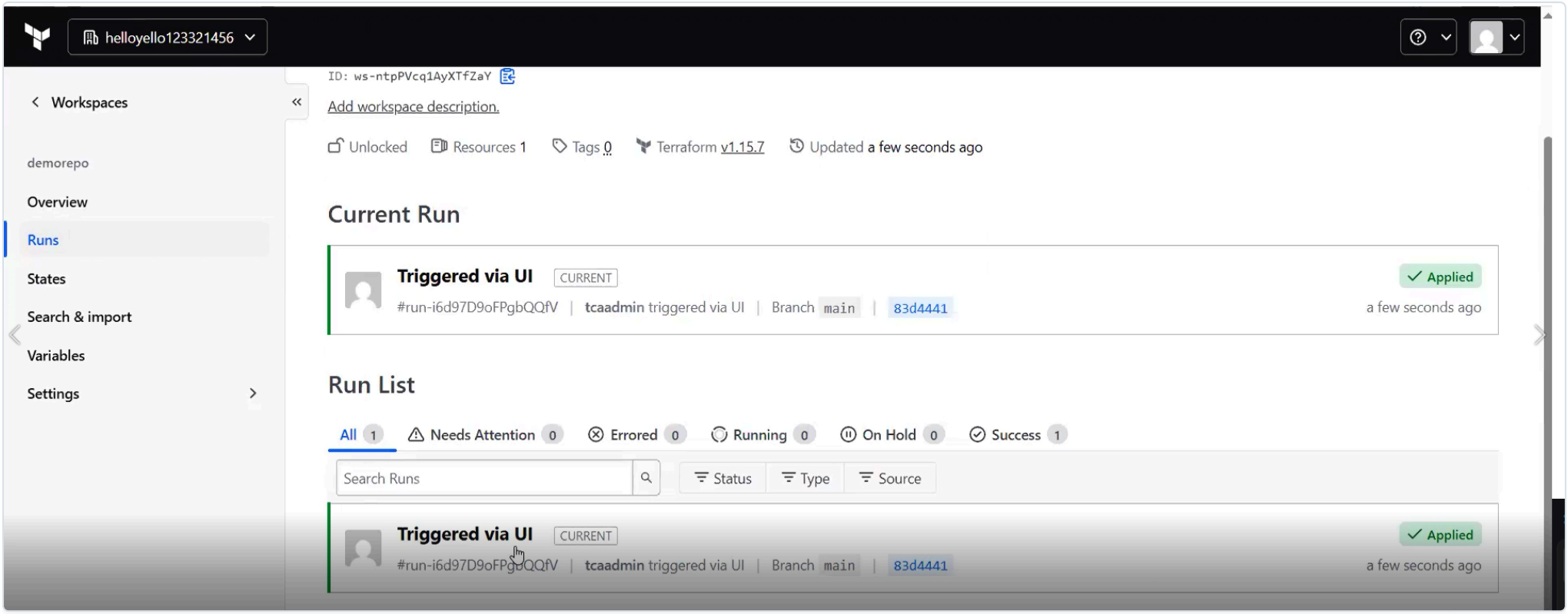

helloyello123321456
Workspaces
demorepo
Overview
Runs
States
Search & import
Variables
Settings
helloyello123321456 / Workspaces / demorepo / Runs
demorepo
Unlocked
Start a new run
Run name
Triggered via UI
Run Type
Plan and apply (standard)
Additional planning options
Start
Cancel
Configure variables
Not configuring variables?

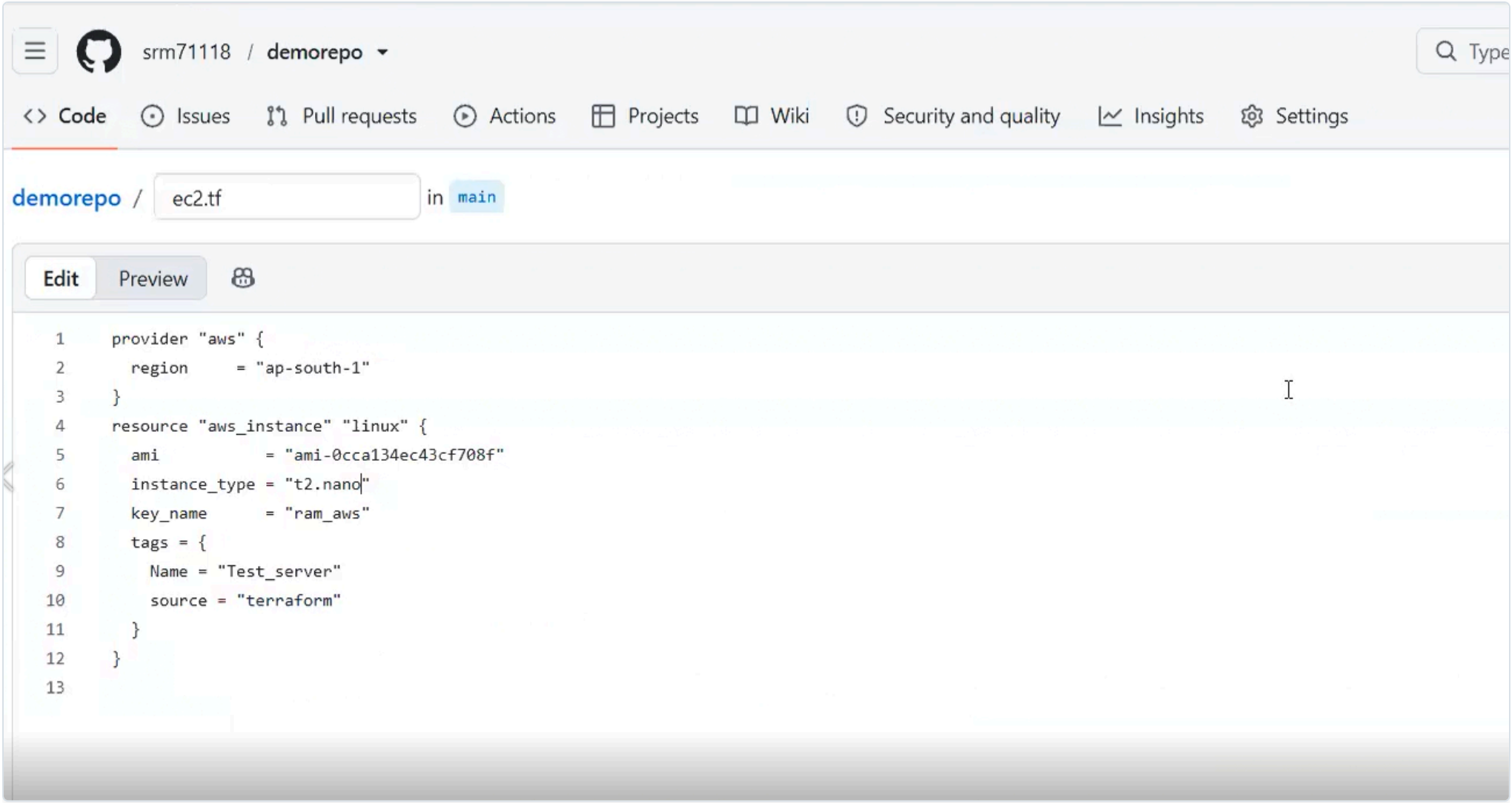

helloyello123321456
Workspaces
demorepo
Overview
Runs
States
Search & import
Variables
Settings
+ host_id : Known after apply
+ host_resource_group_arn : Known after apply
+ iam_instance_profile : Known after apply
+ id : Known after apply
+ instance_initiated_shutdown_behavior : Known after apply
+ instance_lifecycle : Known after apply
+ instance_state : Known after apply
+ instance_type : "t2.micro"
+ ipv6_address_count : Known after apply
+ ipv6_addresses : Known after apply
+ key_name : "ram_aws"
+ monitoring : Known after apply
+ outpost_arn : Known after apply
+ password_data : Known after apply
+ placement_group : Known after apply
+ placement_group_id : Known after apply
+ placement_partition_number : Known after apply
+ primary_network_interface_id : Known after apply
+ private_dns : Known after apply
+ private_ip : Known after apply
+ public_dns : Known after apply

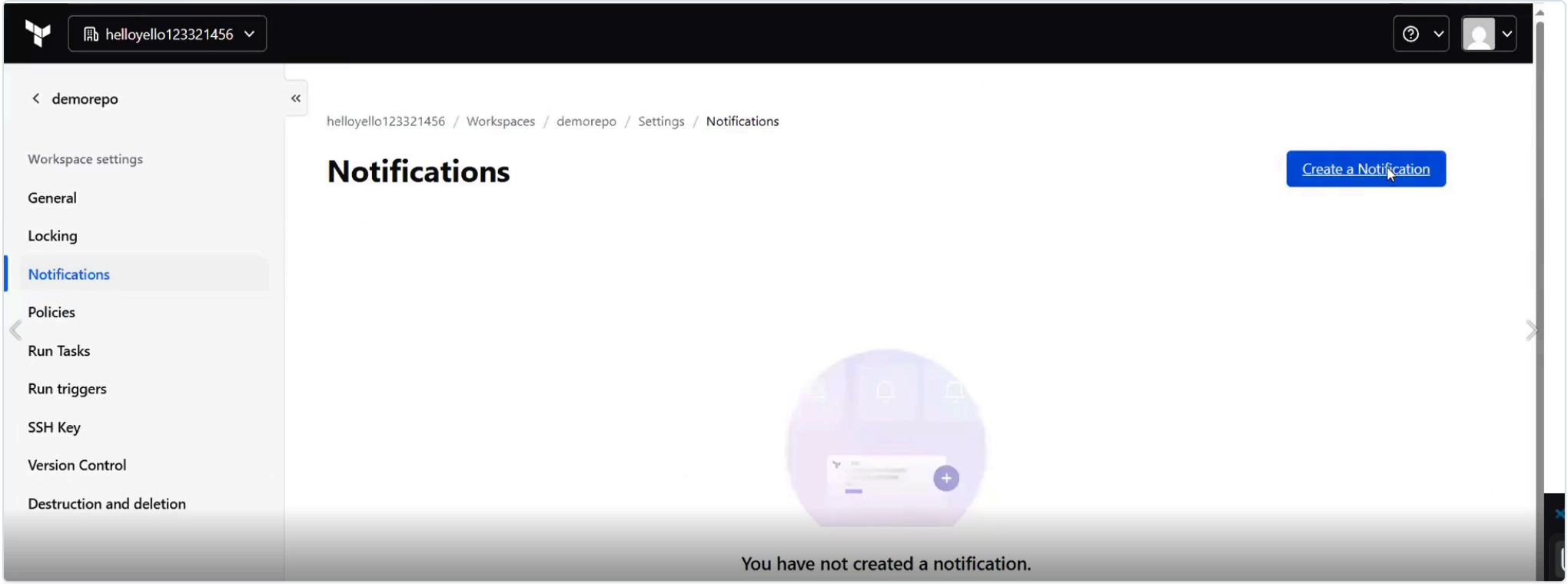

Download Sentinel mocks
Sentinel mocks can be used for testing your Sentinel policies
Apply pending
Confirm & apply
Add a comment to explain this action.
ok go with this plan...

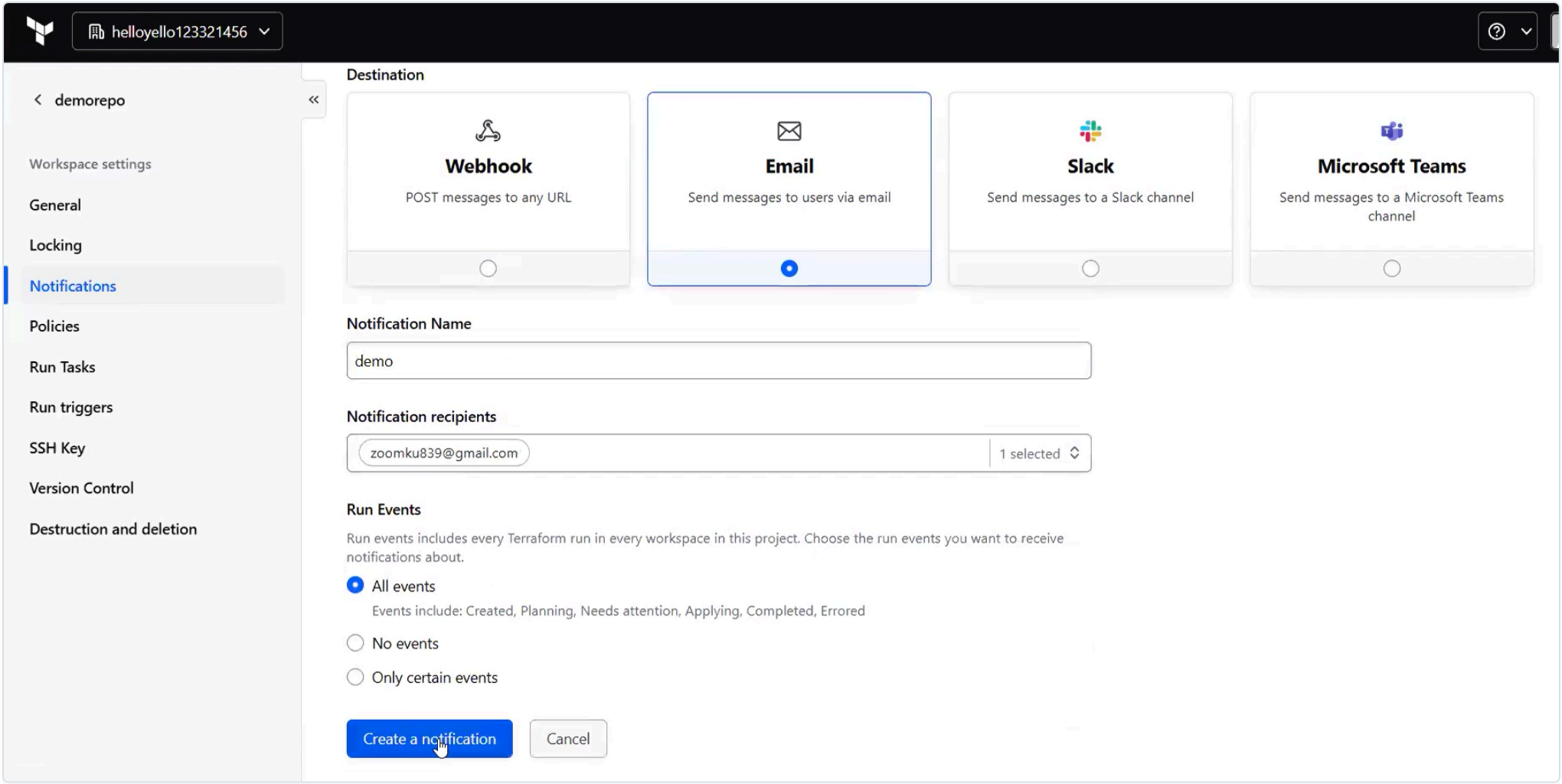

Search
[Alt+S]
Asia Pacific (Mumbai)
AR Network Nuts Pvt Ltd (671609384939)
ram
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Instances (1/1) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Test_server
i-0ee11545846aabf1e
Running
t2.micro
Initializing
ap-south-1b
Unselect instance: Test_server
i-0ee11545846aabf1e (Test_server)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0ee11545846aabf1e
Public IPv4 address
13.235.78.6 | open address
Private IPv4 addresses
172.31.14.110
IPv6 address
Instance state
Running
Public DNS

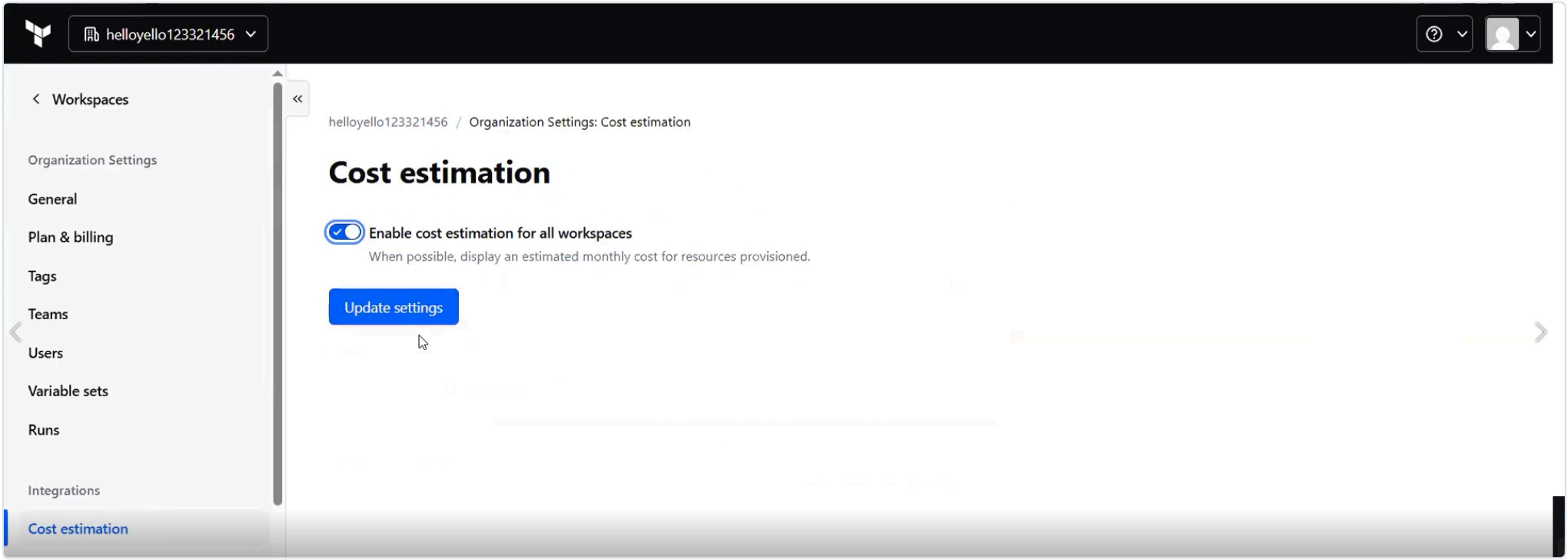

helloyello123321456
Workspaces
demorepo
Overview
Runs
States
Search & import
Variables
Settings
helloyello123321456 / Workspaces / demorepo / States
demorepo
Lock
+ New run
ID: ws-ntpPVcq1AyXTfZaY
Add workspace description.
Unlocked
Resources 1
Tags 0
Terraform v1.15.7
Updated a few seconds ago
Triggered via UI
#sv-kqd1egDe7cKjzRU1 | tcaadmin triggered from Terraform a few seconds ago | #run-i6d97D9oFPgbQQfV | 83d4441

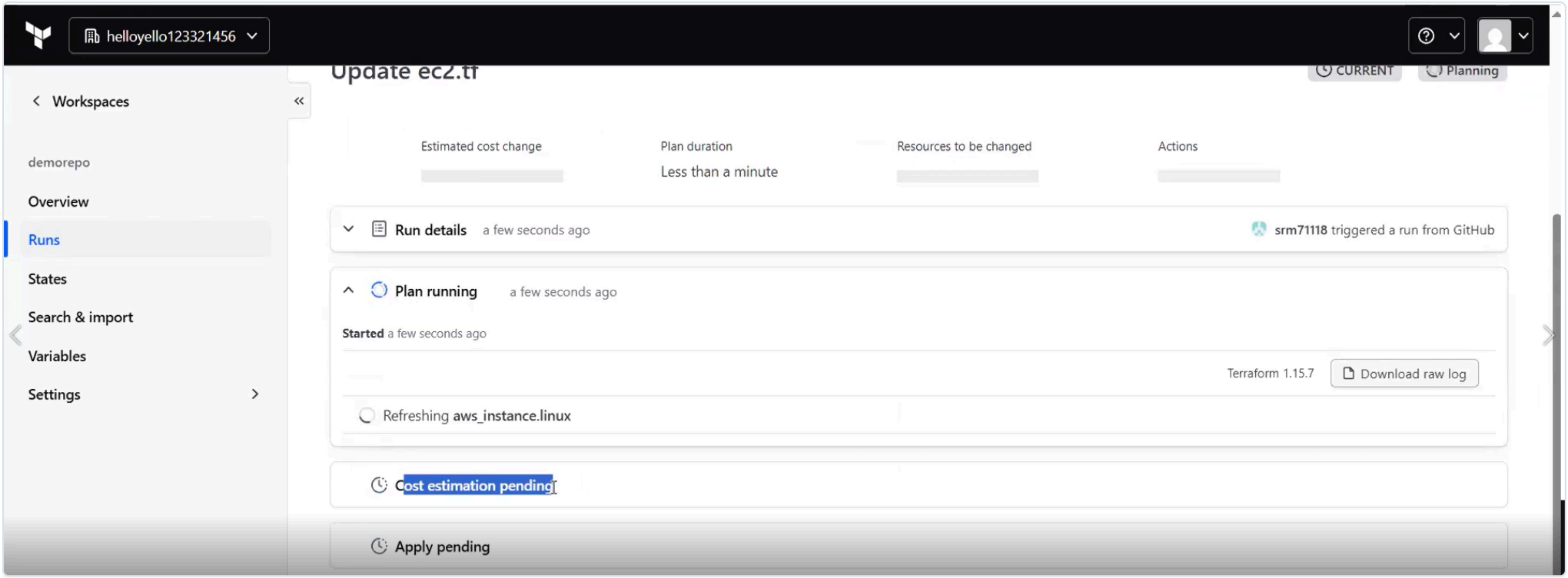

helloyello123321456
Workspaces
ID: ws-ntpPVcq1AyXTfZaY
Add workspace description.
demorepo
Overview
Runs
States
Search & import
Variables
Settings
Unlocked
Resources 1
Tags 0
Terraform v1.15.7
Updated a few seconds ago
Current Run
Triggered via UI
CURRENT
Applied
#run-i6d97D9oFPgbQQfV | tcaadmin triggered via UI | Branch main | 83d4441
a few seconds ago
Run List
All 1
Needs Attention 0
Errored 0
Running 0
On Hold 0
Success 1
Search Runs
Status
Type
Source
Triggered via UI
CURRENT
Applied
#run-i6d97D9oFPgbQQfV | tcaadmin triggered via UI | Branch main | 83d4441
a few seconds ago

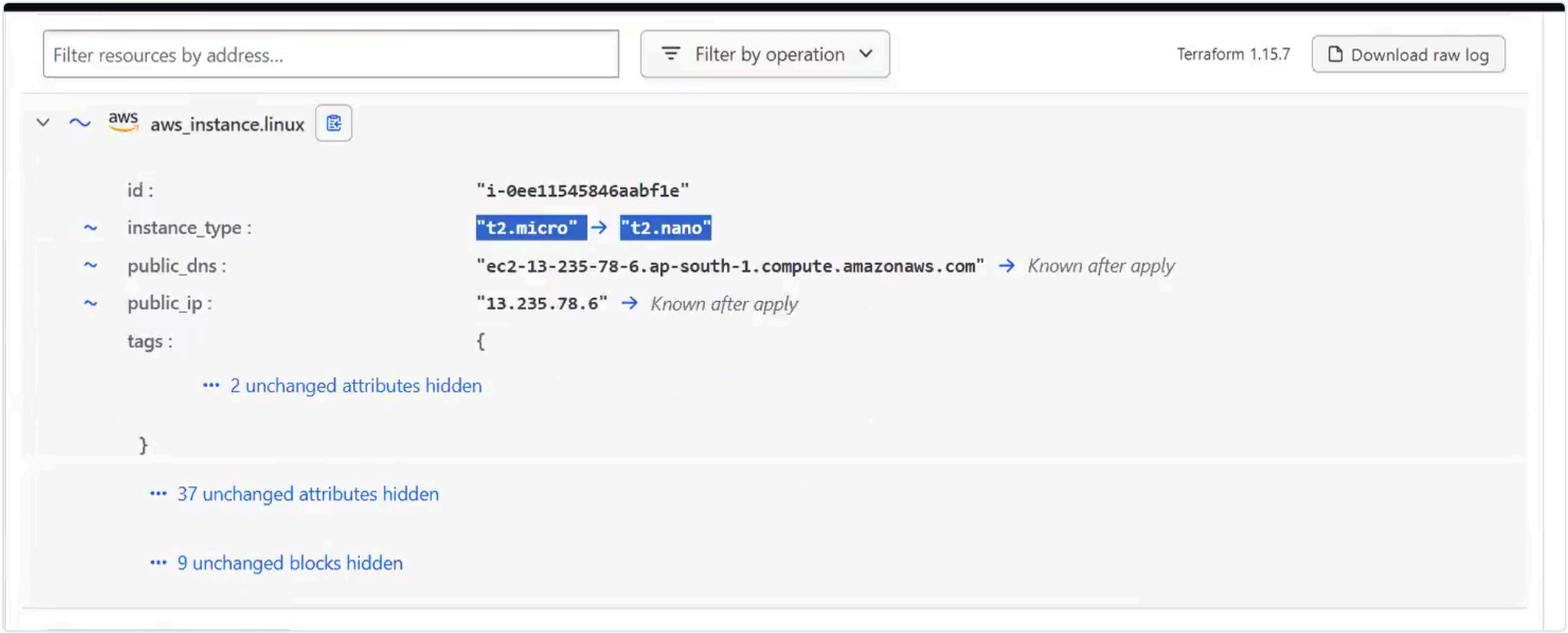

srm71118 / demorepo
Type
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
demorepo / ec2.tf in main
Edit
Preview
provider "aws" {
region = "ap-south-1"
}
resource "aws_instance" "linux" {
ami = "ami-0cca134ec43cf708f"
instance_type = "t2.nano"
key_name = "ram_aws"
tags = {
Name = "Test_server"
source = "terraform"
}
}

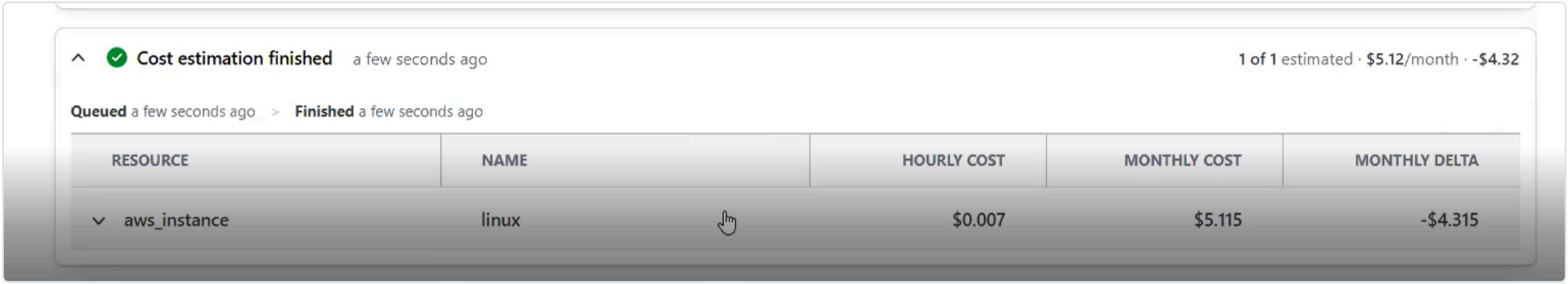

helloyello123321456
demorepo
Workspace settings
General
Locking
Notifications
Policies
Run Tasks
Run triggers
SSH Key
Version Control
Destruction and deletion
helloyello123321456 / Workspaces / demorepo / Settings / Notifications
Notifications
Create a Notification
You have not created a notification.

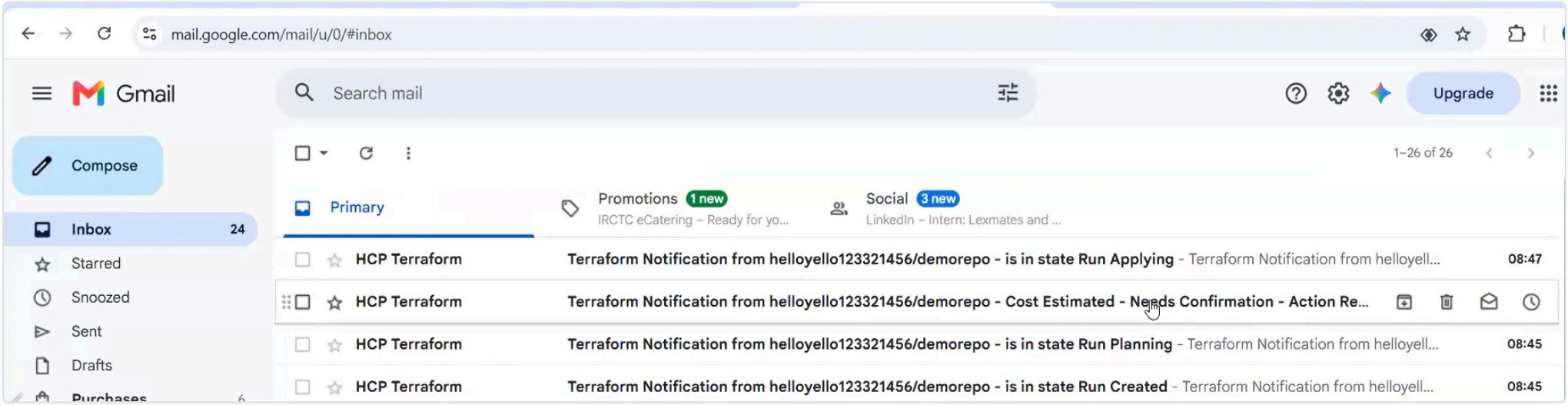

helloyello123321456
demorepo
Workspace settings
General
Locking
Notifications
Policies
Run Tasks
Run triggers
SSH Key
Version Control
Destruction and deletion
Destination
Webhook
POST messages to any URL
Email
Send messages to users via email
Slack
Send messages to a Slack channel
Microsoft Teams
Send messages to a Microsoft Teams channel
Notification Name
demo
Notification recipients
zoomku839@gmail.com
1 selected
Run Events
Run events includes every Terraform run in every workspace in this project. Choose the run events you want to receive notifications about.
All events
Events include: Created, Planning, Needs attention, Applying, Completed, Errored
No events
Only certain events
Create a notification
Cancel

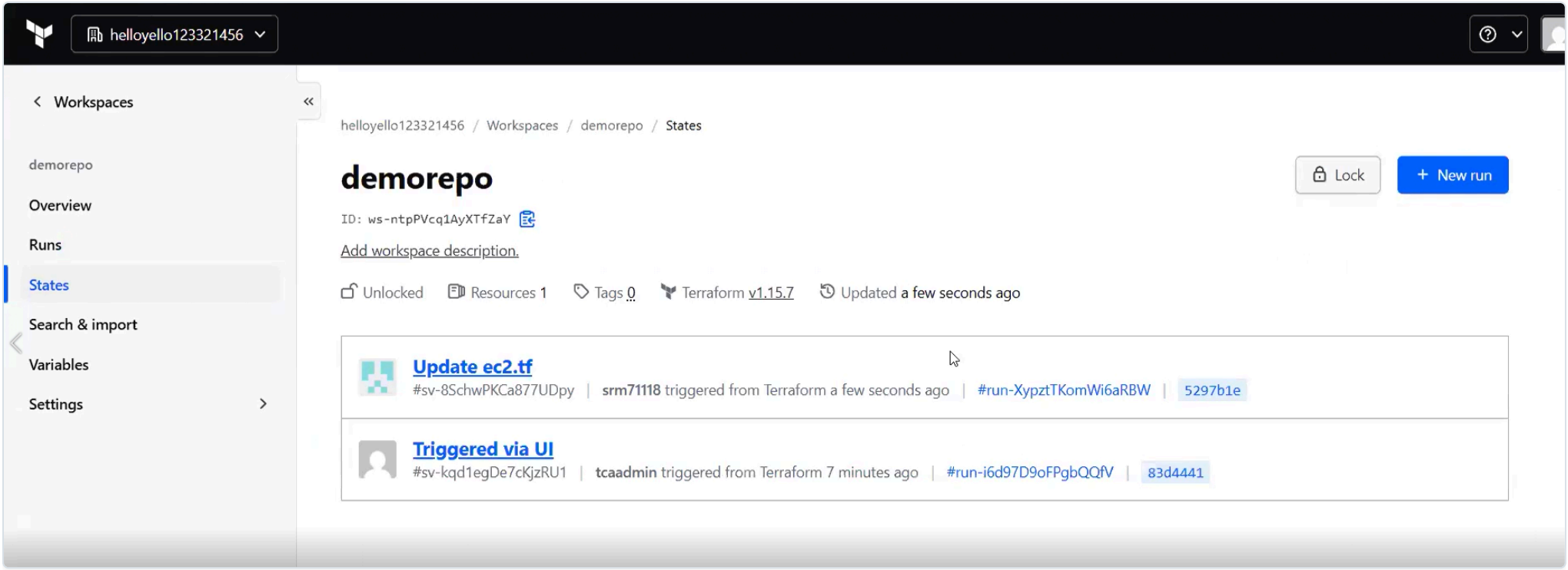

helloyello123321456
Workspaces
Organization Settings
General
Plan & billing
Tags
Teams
Users
Variable sets
Runs
Integrations
Cost estimation
helloyello123321456 / Organization Settings: Cost estimation
Cost estimation
Enable cost estimation for all workspaces
When possible, display an estimated monthly cost for resources provisioned.
Update settings


helloyello123321456
Workspaces
demorepo
Overview
Runs
States
Search & import
Variables
Settings
Update ec2.tf
CURRENT
Planning
Estimated cost change
Plan duration
Less than a minute
Resources to be changed
Actions
Run details a few seconds ago
srm71118 triggered a run from GitHub
Plan running a few seconds ago
Started a few seconds ago
Terraform 1.15.7
Download raw log
Refreshing aws_instance.linux
Cost estimation pending
Apply pending


Filter resources by address...
Filter by operation
Terraform 1.15.7
Download raw log
aws_instance.linux
id : "i-0ee11545846aabf1e"
instance_type : "t2.micro" → "t2.nano"
public_dns : "ec2-13-235-78-6.ap-south-1.compute.amazonaws.com" → Known after apply
public_ip : "13.235.78.6" → Known after apply
tags : {
2 unchanged attributes hidden
}
37 unchanged attributes hidden
9 unchanged blocks hidden


Cost estimation finished a few seconds ago
1 of 1 estimated · $5.12/month · -$4.32
Queued a few seconds ago > Finished a few seconds ago
RESOURCE
NAME
HOURLY COST
MONTHLY COST
MONTHLY DELTA
aws_instance
linux
$0.007
$5.115
-$4.315


mail.google.com/mail/u/0/#inbox
Gmail
Search mail
Upgrade
Compose
Inbox 24
Starred
Snoozed
Sent
Drafts
1–26 of 26
Primary
Promotions 1 new
IRCTC eCatering – Ready for yo...
Social 3 new
LinkedIn – Intern: Lexmates and ...
HCP Terraform Terraform Notification from helloyello123321456/demorepo - is in state Run Applying - Terraform Notification from helloyell... 08:47
HCP Terraform Terraform Notification from helloyello123321456/demorepo - Cost Estimated - Needs Confirmation - Action Re... 08:47
HCP Terraform Terraform Notification from helloyello123321456/demorepo - is in state Run Planning - Terraform Notification from helloyell... 08:45
HCP Terraform Terraform Notification from helloyello123321456/demorepo - is in state Run Created - Terraform Notification from helloyell... 08:45


helloyello123321456
Workspaces
demorepo
Overview
Runs
States
Search & import
Variables
Settings
helloyello123321456 / Workspaces / demorepo / States
demorepo
Lock
New run
ID: ws-ntpPVcq1AyXTfZaY
Add workspace description.
Unlocked
Resources 1
Tags 0
Terraform v1.15.7
Updated a few seconds ago
Update ec2.tf
#sv-8SchwPKCa877UDpy | srm71118 triggered from Terraform a few seconds ago | #run-XypztTKomWi6aRBW | 5297b1e
Triggered via UI
#sv-kqd1egDe7cKjzRU1 | tcaadmin triggered from Terraform 7 minutes ago | #run-i6d97D9oFPgbQQfV | 83d4441